

“Tête-à-tête 22”: Mediating Intimate Relationship through Expressive Digital Interaction

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Figure 1: A Motion Interactive Kinetic Installation, “Tête-à-tête 22”, is placed in a daily setting. It is a set of two sideways tilting chairs connected by a mechanical linkage. Initially, a brake prevents the chairs from tilting, but the seatmates become increasingly dependent on each other to keep their balance when the brake is slowly released, like a relationship.

ABSTRACT

The interactive installation “Tête-à-tête 22” is a conversational sitting object to explore the intimate dependency in relationships that are difficult to achieve remotely. Using a pair of kinetic chairs that gradually become unstable, the installation requires its seatmates work as a team to maintain their balance. The creative technology used in “Tête-à-tête 22” facilitates human-human connections by evoking a sensual experience in intriguing, reflective and expressive ways. Ultimately, the strenuous physical interaction aims to

produce a sense of togetherness in everyday life, by constructing a digital entity with a symbolic meaning, an experiential possibility, and a functional role.

CCS CONCEPTS

• Human-centered computing → Interaction design.

KEYWORDS

Affective Computing, Expressive Digital Interaction, Interactive Art, Meta Materials

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1 INTRODUCTION

Due to the pandemic, we are spending more time in front of a computer and phone to communicate with others. While technology provides greater safety and flexibility in interacting with each other, we now have fewer opportunities to interact physically. In addition, to remove distractions and ensure a better connection with the person behind the screen, a private room and individual setting is preferred. The results are a lack of sharing experiences in the same place at the same time and attention paid to bodily cues between people. This loss of proximity has left many people feeling isolated, distanced, and disengaged with even the closest of family members and co-workers. These feelings of being emotionally and sensually absent led us to imagine and create “Tête-à-Tête 22”.

1.1 Bring Felt-Experience Into Everyday Life

Digital objects play a significant role in our daily lives in ways that go far beyond any straightforward notion of ‘use’. They continually shape lived experience and establish key frameworks for how people think, act, and live. But, the way we are interacting with daily digital technology today lacks the nuance needed for complex socio-emotive matters, such as interaction intimacy, affinity, conflicting emotional resonances, and ambiguity. There is thus an opening for design research to explore more thoughtful and creative approaches to creating digital objects. Through “Tête-à-tête 22”, we seek sensual possibilities for reconnection that can be achieved by interacting with a digital entity. This is based on the concept of “felt experience” in technology which is a way to view technology as an experience that enables opening of the sensual, emotional, volitional and dialogically imaginative aspects [2]. We strive for opening a creative digital entity’s role to mediate people’s emotional interdependence by enabling people bodily to feel self and another person, and that opens further opportunities to express to each other. Thus, the notion of “feltness-technology” [2] can be utilized for bringing more human values such as authenticity and relationships into daily life.

1.2 Creating Intimacy Through Strenuous Interaction

Throughout the design process, we wanted both aesthetic and interaction components to capture a deep sense of “togetherness” between people. Hence the title “Tête-à-tête”, defined as “a private conversation between two people” or “a face-to-face meeting” [4]. The distinctive two-seaters style of sofas (a face-to-face orientation with a shared armrest) have been designed in this context. As a twist, we designed a side-by-side sitting version of “Tête-à-tête 22” without any armrest to open an opportunity for people to support each other’s arms. Aesthetically, the piece consists of two kinetic chairs with expressive backrests, and the seating arrangement enables two people to have closer physical interaction by seeing the same view at the same time. In the era of internet-based interaction, which requires the constant screen-to-screen setup, we observed that these two orientations were sorely missed nowadays in non-collocated relationships.

The two chairs, as shown in Figure 1, are set up on one frame and connected by a mechanical linkage that couples the sideways rotation of both chairs, as shown in Figure 2. At first, when people



Figure 2: The chairs are tilting left and right when a pair of people are seated.

sit down, a brake fixes the linkage and makes the chairs static. This is just like when people meet for the first time in real life, and everything seems to go smoothly. Then, the brake is slowly released. The result is that the chairs will lose stability if either party tries to adjust posture or sitting position. In order to not fall off the chairs, the sitters are encouraged to physically hold each other to maintain their balance. Whether through words or nonverbal communications, they must synchronize their body movements, which often involves touching and grabbing the other’s hands, arms or legs. Along with the strenuous interaction that induces people to help one another, the process may create an emotional deeper bond between the participants. Thus, interacting with the restless digital furniture becomes a playful, and yet uncomfortable and exertive platform that facilitates further mimicking intimacy in people’s relationships. In this way, technology can be evolved more creatively by opening new ways to explore its value and role along with actively co-shaping people’s beings in the world, rather than serving people as a mere background [3]. In this post COVID world, people are wary of being close and touching, even if they miss it. “Tête-à-tête” will be one of the conjunctions that physically mediates people to regain intimate dependency.

2 DESIGN

2.1 The Chair

The chairs, shown in Figure 3, are made out of two layers of steel sheet, laser cut to create both the chairs’ contour and the decorative piercing. Then, each layer is carefully bent as a long organic curve to form three parts: a sitting pad, a leg, and a footrest. The parts are expertly welded together top and bottom frames with stitching another metal connector to create an extremely thin and strong



Figure 3: The chair is laser cut out of two layers of sheet metal (1mm), bended and welded together (left). The hand-crafted leather component is elaborated onto a metal chair (centre and right).

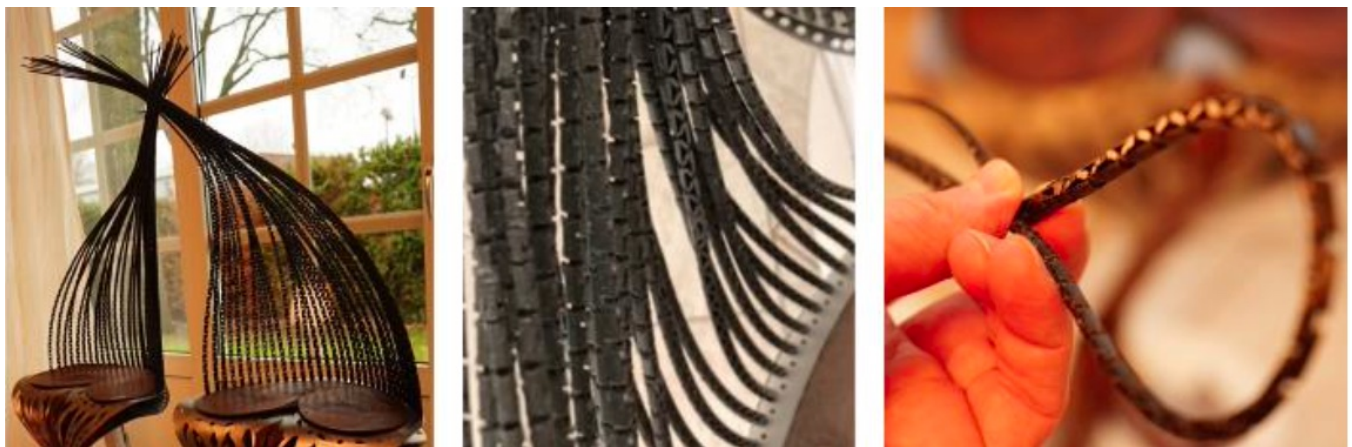


Figure 4: The hairs are 3D printed (FDM/PLA) and tapered; the meta material structure provides the flexibility.

structure. As a second feature, the metal chair frames are paired with leather material that is created by pyrography (engraving rich tones and patterns on the leather by using heat), bringing hand-drawn details, shown in Figure 3 centre and right.

Mechanically, both chairs rotate just under the seat on the sagittal axis, tilting the person sideways. A simple mechanical linkage connects both chairs together with a metal bar and synchronizes their rotation. The rotation is entirely powered by the balance of the seatmates. A brake, modeled after a bicycle rim brake, only restricts rotation by clamping the linkage bar to the structure and transfers the responsibility of stability from the installation to the seatmates when it is released.

2.2 The Backrest

The backrest is built like strands of hair and the two back seats are joined together like a bundle of hair, shown in Figure 4. The strings are 3D printed out of PLA, and a total of 62 hair models are connected to the chairs with the bolts and nuts. Each hair structure consists of a long mechanical meta material that is built by three segments, and it is designed to be bendable and stretched only side to side, shown in Figure 4, right. When the chairs lose their balance and swing, the backrests between the two chairs will create pull and push motions from one middle cleavage. It looks as if two bundles of hairs are simultaneously yanking each other. The human

movements are dramatically developed on those unstable chairs, emphasizing that two human beings are connected. After the pair of sitters are able to sit side by side with a good balance, the hair motion stops, and the backrests are returned to being calm and static again. At that point, it'll be more likely that two people will agree to support (touch) each other's arms to get balanced together, much like two backrests joined into one.

2.3 Human-Human Communication Through Expressive Digital Interaction

Expression is the important key to being able to communicate and connect in a human relationship. Thus, "Tête-à-tête 22" explores how expressiveness can be enacted for both digital entity and human. The artifact symbolizes human relationships as an expressive design piece. The motion of two interactive chairs illustrates the complexities of building and sustaining human relationships: the ups and downs unpredictable waves and the need for cooperation in order to co-exist, shown in Figure 5. Furthermore, the metaphor imparts a greater sense of physical experience when two people are seated on it, it elicits people to converse with one another while learning how to align their body movements and minds throughout the uneasy and uncomfortable interaction.

While the kinetic motion expresses how two entities are connected and affect each other, the subsequent interaction enables



Figure 5: “Tête-à-tête 22”, The interactive back seats.

two people to reflect on their struggles, excitements, appreciation, and annoyance about being together. In specific, the mechanism enables one to physically experience the other’s movement and what it’s like to feel the other’s presence. Moreover, it provides an opportunity to recognize how the other is going to act, react accordingly, and work in tandem to achieve the requisite balance. Lastly, “Tête-à-tête 22” may prompt further expression and contemplation between the sitters about their intentions, reactions, and feelings. Hallnäs and Redström [1] stated that technology could be played as an expressive tool to invite reflection, and it’s reflective in its expression between human and digital entities. In summary, “Tête-à-tête 22” is a compelling example of an expressive digital tool that opens bodily communication, encourages interaction, and invites

reflection. In this way, Human-Computer Interaction can produce aesthetic interaction aiming to foster Human-Human interaction.

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